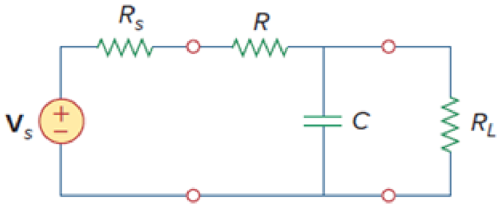
Chapter 14, Problem 102CP

Problem

Practical *RC* filter design should allow for source and load resistances as shown in Fig. 14.110. Let $R = 4$ kΩ and $C = 40$-nF. Obtain the cutoff frequency when:

(a) $Rs = 0$, $RL = \infty$,

(b) $Rs = 1$ kΩ, $RL = 5$ kΩ.



Step-by-step solution

Step 1 of 2

(a) When $RS = 0$ and $\text{RL} = \infty$, we have a Low pass filter

$$\omega_c = 2\pi fC = \frac{1}{RC}$$

$$fC = \frac{1}{2\pi RC} = \frac{1}{(2\pi)(4\times10^3)(40\times10^{-9})}$$

$$= 994.7 H_Z$$

Step 2 of 2

(b) We obtain R_{TH} across the capacitor

$$R_{TH} = R_L \,||\, (R + R_S)$$

$$R_{TH} = 5 \,||\, (4+1) = 2.5\,K\Omega$$

$$fC = \frac{1}{2\pi R_{TH} C} = \frac{1}{(2\pi)(2.5\times10^3)(40\times10^{-9})}$$

$$fC = 1.59\,KH_Z$$